

STA 5635 HW 5

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Problem 1

- (1a) Loss function: $L(\mathbf{w}) = \frac{1}{N} \sum_{i=1}^N f(y_i \mathbf{x}_i^T \mathbf{w}) + s \|\mathbf{w}\|_2^2$, where $f(x) = \max(0, 1 - x)$. I do not believe that the derivative of $f(x)$ exists, at least not cleanly. The portion of the loss consisting of the average of the function only contributes if the function doesn't return 0. In other words:

$$f(y_i \mathbf{x}_i^T \mathbf{w}) = \begin{cases} 0 & y_i \mathbf{x}_i^T \mathbf{w} \geq 1 \\ 1 - y_i \mathbf{x}_i^T \mathbf{w} & y_i \mathbf{x}_i^T \mathbf{w} < 1 \end{cases}$$

So for a point where $y_i \mathbf{x}_i^T \mathbf{w} < 1$, the gradient is $\frac{\partial}{\partial \mathbf{w}}(1 - y_i \mathbf{x}_i^T \mathbf{w}) = -y_i \mathbf{x}_i^T$. For a points where $y_i \mathbf{x}_i^T \mathbf{w} \geq 1$, the gradient is $\frac{\partial}{\partial \mathbf{w}}(0) = 0$. All together, the total gradient is:

$$\nabla_{\mathbf{w}} = \frac{-1}{N} \sum_{i: (y_i \mathbf{x}_i^T \mathbf{w}) < 1} (y_i \mathbf{x}_i^T) + 2s \mathbf{w}$$

We plot the training loss vs. iteration number in table 1.

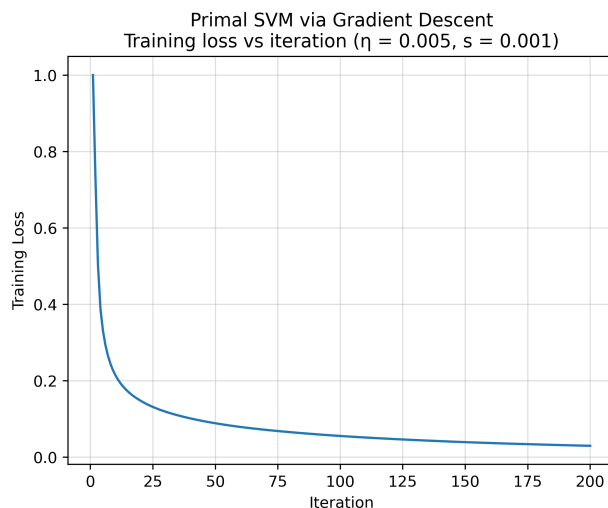


Figure 1: Primal SVM Training Loss vs. Iterations

The table outlining the training and testing misclassification errors is shown in figure 1:

Method	Train Misclass	Test Misclass
Primal SVM	0.006333	0.019000

Table 1: Train and Test Misclassification Errors, Primal SVM via Gradient Descent.

Method	Train Misclass	Test Misclass	Num. Support Vectors
Linear SVM	0.000000	0.019000	1236

Table 2: Train, Test Misclassification Errors and Number of Support Vectors, Linear SVM, C=1

- (1b) Use SVM package to train linear SVM, C=1. We can report the training errors and number of support vectors in table 2: Compared to part a, the test misclassification error is the same, but the train misclassification error is slightly lower.
- (1c) Use SVM package to train a SVM with polynomial kernel of degree 2. We can report the training error and number of support vectors in table 3: Compared to part b, the

Method	Train Misclass	Test Misclass	Num. Support Vectors
Quadratic SVM	0.000167	0.025000	4744

Table 3: Train, Test Misclassification Errors and Number of Support Vectors, Quadratic SVM, C=1

number of support vectors is almost four times as great.

- (1d) Train an SVM classifier with C=1 and RBF kernels $\gamma = 10^{-i}, i \in \{1, 2, \dots, 6\}$. We can plot the train and test misclassification errors vs γ in figure 2: We can also plot the number of support vectors vs γ , with a dashed line showing the number of support vectors in part b in figure 3:
- (1e) asdf

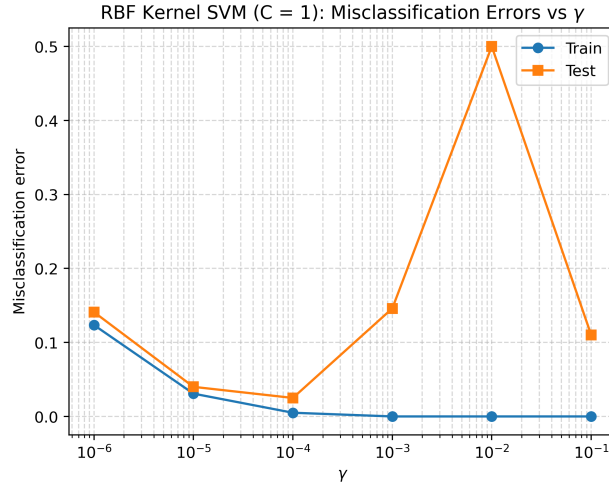


Figure 2: SVM Classifier, $C=1$, Train and Test Misclassification Errors vs γ .

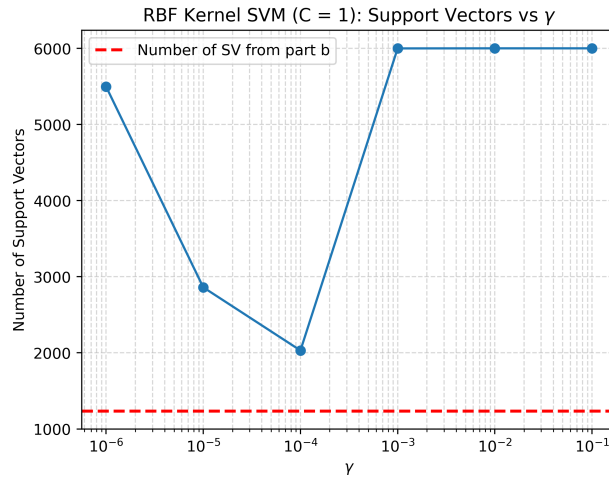


Figure 3: SVM Classifier, $C=1$, Number of Support Vectors vs γ . Dashed line indicates number of support vectors from part b (quadratic SVM).